

Master Seminar Talk

Overview and current developments

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1 Introduction

2 Experimental Background

- The optical centrifuge
- Droplets and Molecules
- Measurement Technique

3 New Data

- General Information
- CS₂ in Droplets
- OCS in Droplets

4 Current Theory

- Spinnability
- Field-free Rotation
- In-Field Rotation

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A view key ideas

- **What are we trying to do?**

- Spinning molecules in superfluid ^4He
- Use different methods to bring energy in this microscopic system
- compare with free molecules in the gas phase

- **Goal of the research**

- figure out properties of superfluidity, such as: "friction" (energy dissipation), interaction with molecules, breakdown of superfluidity

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Animation of an accelerating optical centrifuge

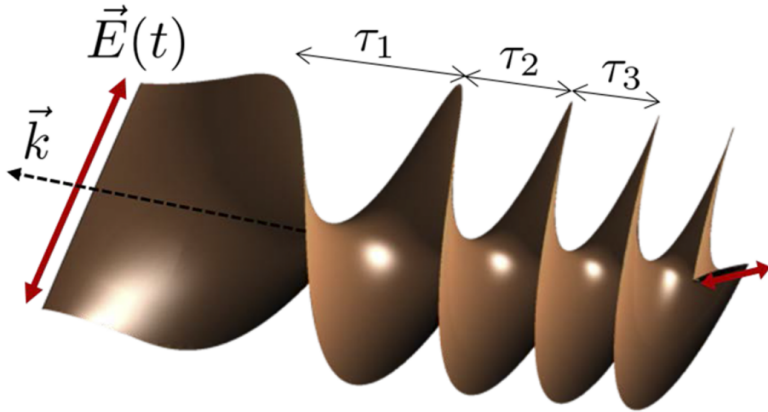


Figure: E-field of an accelerating optical centrifuge¹

¹MacPhail-Bartley, Wasserman, et al. 2020a.

How to make an optical centrifuge

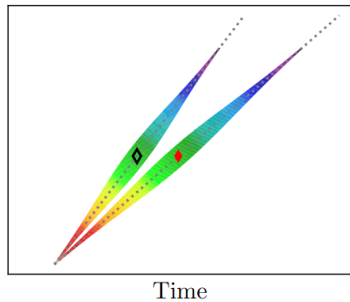


Figure: Chirped pulses used to create the accelerating rotating electric field^a

- Take two pulses with **equal amplitudes** and **opposite circular polarizations**: one right-circular (RCP), one left-circular (LCP).
- At any instant, **RCP + LCP of equal strength** adds up to a **linearly polarized** electric field.
- The **only thing that matters** for the *direction* of that linear polarization is the **relative optical phase** between the two components.
- If that relative phase changes in time, the resulting linear polarization axis **rotates in time**.

^aWang et al. 2025.

^4He nanodroplets & field coupling to linear molecules

- **^4He nanodroplets:** formed by supersonic expansion through a cryogenic nozzle; cool to $\sim 0.4\text{ K}$ and are typically **superfluid**.
- **Size:** usually $N \sim 10^3\text{--}10^6$ **He atoms** $\Rightarrow$ radius $10 - 100\text{ nm}$ (source-condition dependent)².
- **Molecule pickup:** droplets capture molecules in a pickup cell and rapidly **cool/thermalize** them.
- **E-field coupling (nonresonant):** alignment via anisotropic polarizability³

$$U(t) \propto -\Delta\alpha E^2(t) \cos^2 \theta, \quad \Delta\alpha = \alpha_{\parallel} - \alpha_{\perp}.$$

- For $\Delta\alpha > 0$ the molecule aligns with the instantaneous polarization axis; if that axis rotates (optical centrifuge), the molecule can be **captured and spun up** to high J .

²Toennies and Vilesov 2004.

³MacPhail-Bartley 2024.

Spinning of NO dimers

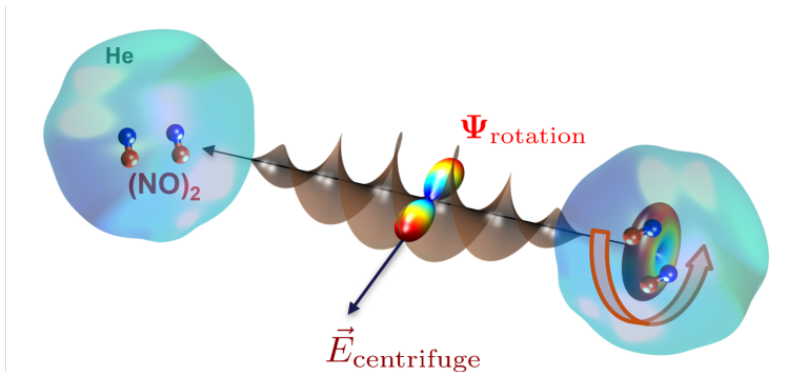


Figure: Spinning of NO dimers in helium nanodroplets with optical centrifuge⁴

⁴Home Page | Ultrafast Coherent Control Group 2026.

Measurement principle: Coulomb explosion + VMI readout

- **Prepare:** Molecules are embedded in ^4He nanodroplets and are aligned/spun up by the pump (e.g. optical centrifuge) at a controlled pump–probe delay.
- **Coulomb explosion (probe):** A short, intense probe pulse removes multiple electrons $\Rightarrow$ the molecule becomes a **highly charged ion** and fragments.
- **Axial recoil:** Fragment ions recoil mainly along the **molecular bond axis** (for suitable channels) $\Rightarrow$ ion emission angles encode the instantaneous molecular orientation.
- **Velocity Map Imaging (VMI):** Electrostatic lenses map the 3D ion velocity distribution onto a 2D detector: ions with the same transverse velocity hit the same detector radius.
- **Detection & analysis:** The detector image (MCP/phosphor + camera) gives **angular distributions** (alignment/orientation) and **kinetic energy release**; repeating vs delay reconstructs the rotational dynamics.

Example of one measurement frame

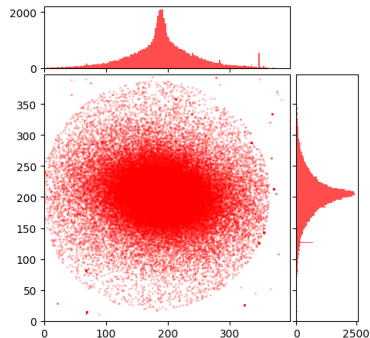


Figure: Detected ions (unprocessed)

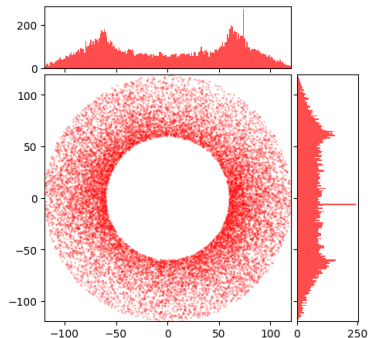


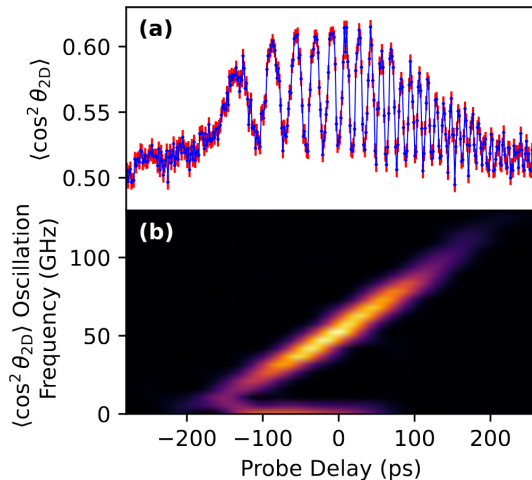
Figure: Detected ions (processed)

- We scan the pump-probe delay

$$\tau = t_{\text{probe}} - t_{\text{pump}}$$

to map the time-dependent rotational dynamics.

Measurement data from a typical scan



- $\cos^2(\Theta)$ data from a typical scan
- centrifuge accelerates from 0 **GHz** to about 120 **GHz**
- central frequency of the pump pulse approx. 50 **GHz**

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General information

- to be certain in our droplet measurements we normally compare gas-jet and droplet experiments
- **Gas-jet**: free rotating molecule at around **10-15K**
- **Droplets**: molecules attach **helium belt** and are cooled down to about **0.4K**

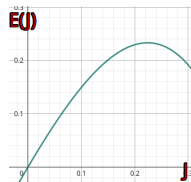


- different **B** and **D** rotational constants
- different thermal **state** distribution
- **interactions** with the helium in droplets

Typical model used to describe the molecule state: **Rigid Rotor with Centrifugal Distortion**

$$E(J) = BJ(J+1) - DJ^2(J+1)^2$$

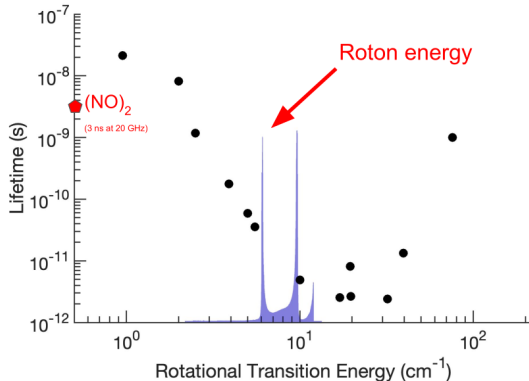
→ the **wall** !!!



Energy dissipation in superfluid helium

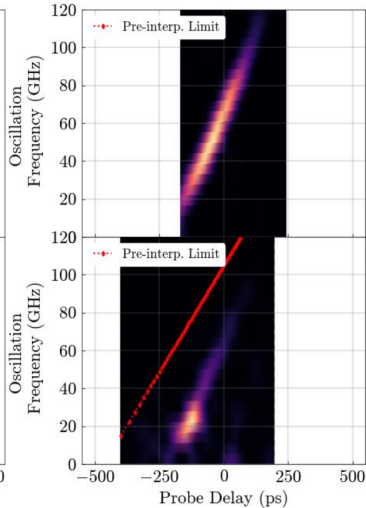
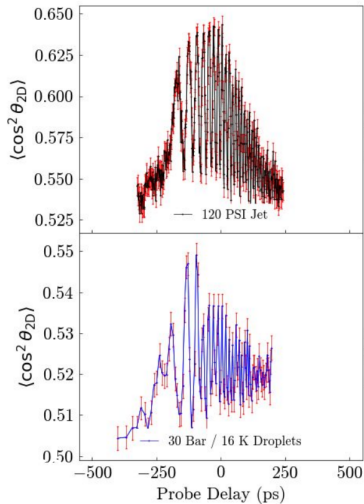
Landau velocity (ideal picture)

- no energy dissipation/ elementary excitation in the superfluid below critical kinetic energy
- "friction" through excitation of Rotons



- only experimentally determined lifetimes
- different measurement methods
- Roton energy around **180 GHz**

CS2 - fast centrifuge

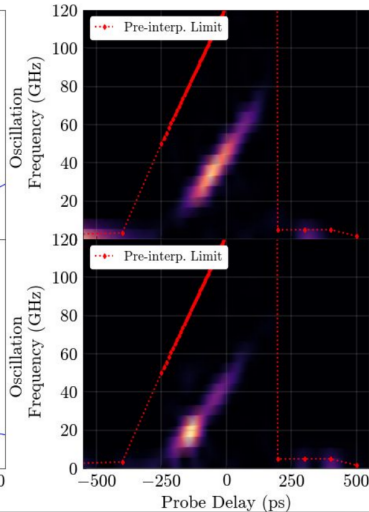
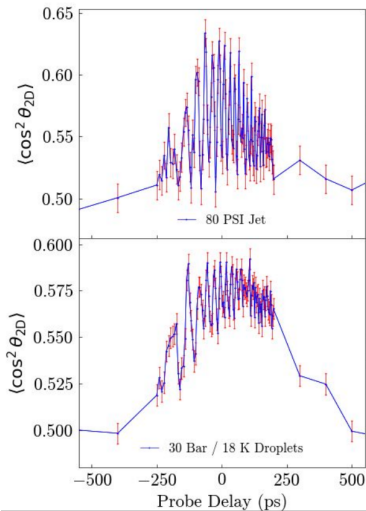


• wall at 20 GHz

New:

- break through wall
- intensity decrease

OCS - slower centrifuge

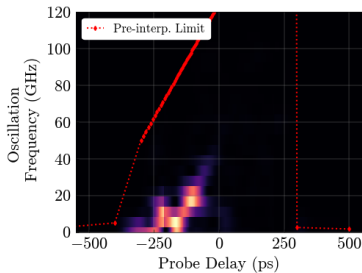
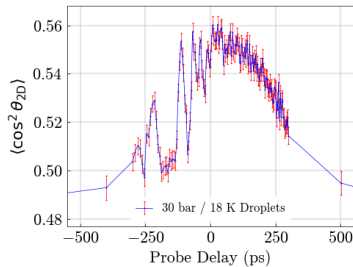
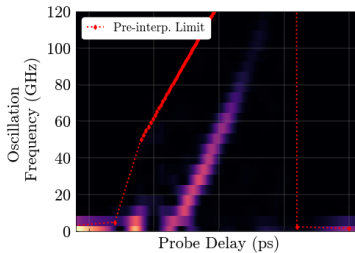
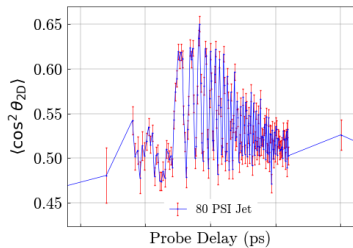


- wall at 38 GHz

New:

- decrease in intensity around the same frequency
- higher baseline

Modelling the in-field dynamics



• wall at 38 GHz

New:

• not following after 20 GHz

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Spinnability in an Optical Centrifuge

For a molecule in a rotating linearly polarized laser field, the key requirement is that the molecule can *adiabatically follow* the rotation of the field. This means that the angular acceleration induced by the field must be larger than the angular acceleration of the rotating polarization⁵.

$$\sigma_0 := \frac{U_0}{\pi I \beta} = \frac{1}{4\pi} \frac{E_0^2}{\beta} \frac{\Delta\alpha}{I} > 1 \quad (1)$$

with

$$U_0 = \frac{1}{4} E_0^2 \Delta\alpha. \quad (2)$$

Meaning of the quantities:

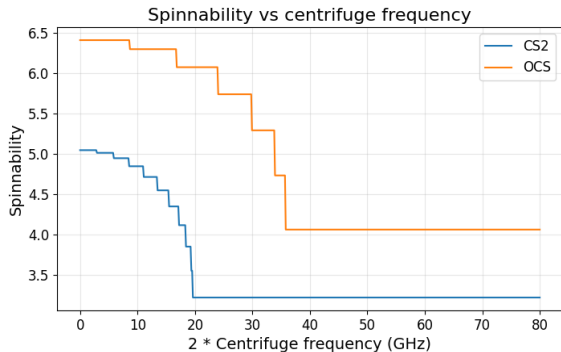
- E_0 : laser field amplitude
- $\Delta\alpha = \alpha_{\parallel} - \alpha_{\perp}$: polarizability anisotropy
- I : relevant moment of inertia of the molecule
- β : chirp rate of the centrifuge field, with $\ddot{\theta}_{\text{pol}} = 2\beta$
- U_0 : depth of the aligning potential

⁵MacPhail-Bartley, Wasserman, et al. 2020b.

OCS vs CS₂ in Theory

Table: Droplet rotational constants and anisotropic polarizabilities for CS₂ and OCS.

Molecule	B (droplet)	D (droplet)	$\Delta\alpha$
CS ₂	730 MHz	1.2 MHz	$8.7 \text{ \AA}^3$
OCS	2.18 GHz	9.5 MHz	$3.7 \text{ \AA}^3$



- does not really explain the different behaviors of the molecules
- model assumes **field-free** states

Hamiltonian for Field-Dressed States

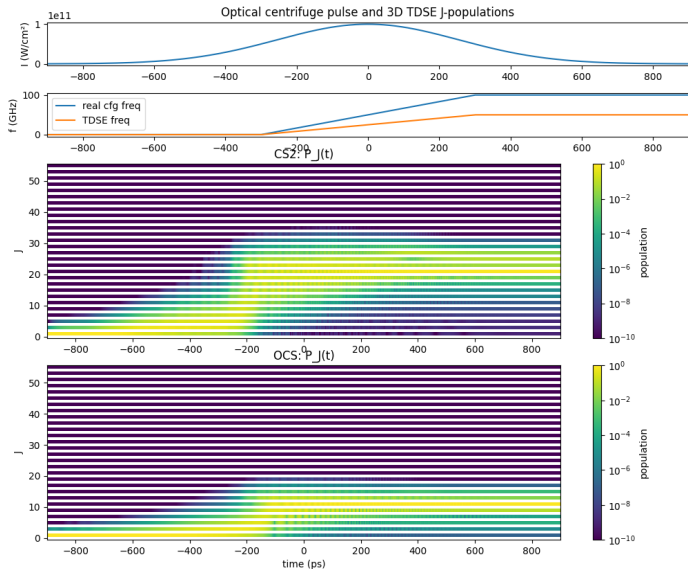
The rotating-frame Hamiltonian

$$\hat{H}_{\text{dressed}}(t) = B\hat{J}^2 - D\hat{J}^4 - U_0(t)\cos^2\theta - \Omega(t)\hat{J}_z. \quad (3)$$

Interpretation:

- $B\hat{J}^2 - D\hat{J}^4$: field-free rotational structure
- $-U_0(t)\cos^2\theta$: aligning interaction with the laser field
- $-\Omega(t)\hat{J}_z$: rotation-frame term favoring co-rotating angular momentum
- The instantaneous eigenstates of $\hat{H}_{\text{dressed}}(t)$ are the **field-dressed states**
- In the adiabatic limit, the molecule follows one of these dressed states as $U_0(t)$ and $\Omega(t)$ evolve

Field Dressed States



Questions?

Thank you for your attention!

-  *Home Page | Ultrafast Coherent Control Group* (Jan. 2026). [Online; accessed 7. Jan. 2026]. URL: <https://coherentcontrol.sites.olt.ubc.ca>.
-  MacPhail-Bartley, Ian (June 2024). “Control of Molecular Rotation in Superfluid Helium”. PhD thesis. Vancouver, BC, Canada: The University of British Columbia.
-  MacPhail-Bartley, Ian, Walter W. Wasserman, et al. (Apr. 2020a). “Laser control of molecular rotation: Expanding the utility of an optical centrifuge”. In: *Review of Scientific Instruments* 91.4. ISSN: 1089-7623. DOI: 10.1063/1.5140358. URL: <http://dx.doi.org/10.1063/1.5140358>.
-  — (Apr. 2020b). “Laser control of molecular rotation: Expanding the utility of an optical centrifuge”. In: *Review of Scientific Instruments* 91.4, p. 045122. ISSN: 0034-6748. DOI: 10.1063/1.5140358. eprint: https://pubs.aip.org/aip/rsi/article-pdf/doi/10.1063/1.5140358/19771927/045122_1_online.pdf. URL: <https://doi.org/10.1063/1.5140358>.
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Wang, Kevin et al. (2025). *An ultraslow optical centrifuge with arbitrarily low rotational acceleration*. arXiv: 2512.20568 [physics.chem-ph]. URL: <https://arxiv.org/abs/2512.20568>.